



OPTIMIZED ROUTE RECOMMENDATION SYSTEM FOR TAXI DRIVERS IN DAEJEON



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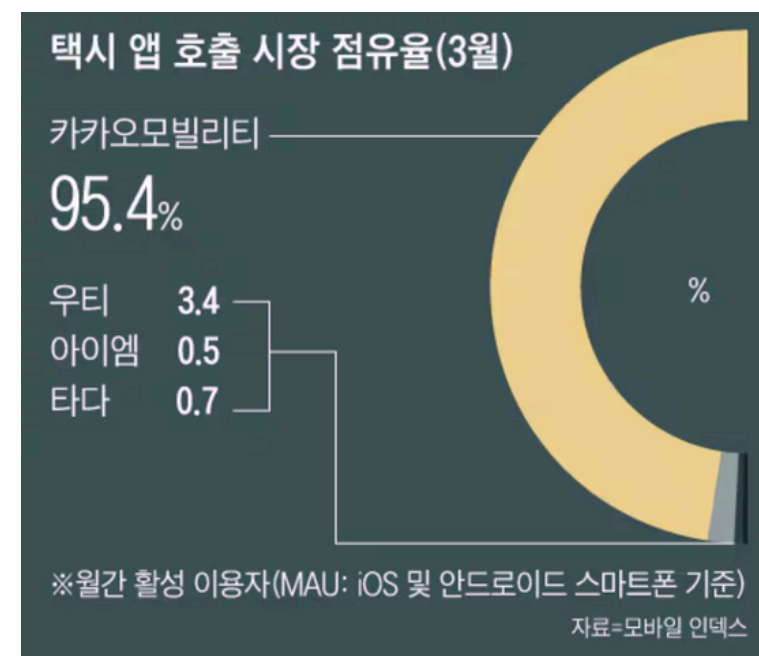
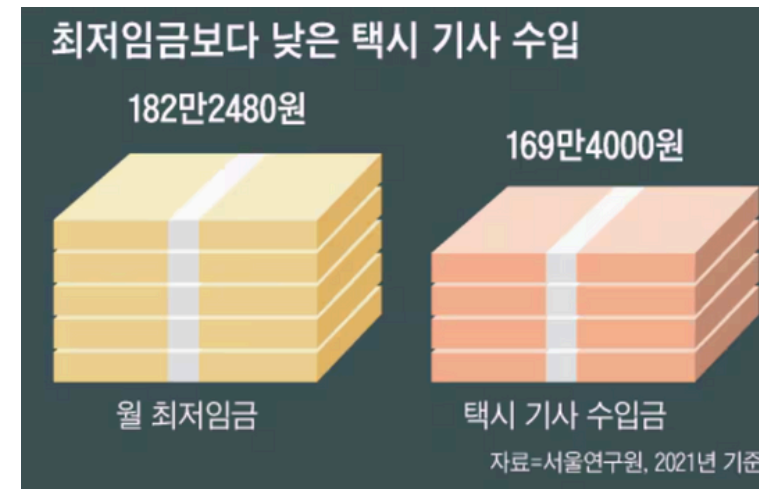
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EXPECTED IMPACT &
FUTURE IMPROVEMENTS

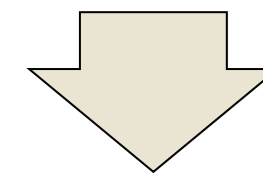


PROBLEM BACKGROUND & GOAL

With the emergence of ride-sharing services and ride-hailing mobile apps, the traditional taxi industry is facing significant challenges.



- Decreased profitability due to **long waiting times and inefficient operations**
- Relying on connection systems such as Company K to secure passengers, and paying **fees** for these services
- **Financial burden on drivers**



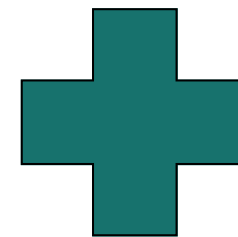
We aim to provide a service that presents **an efficient driving strategy** to generate better earnings!

DATA UTILIZATION & PREPROCESSING WORKFLOW

Operation data of 205 taxis in Daejeon

(One year of operation records from April 2023 to March 2024)

Basic date filtering and removal of missing values



Korean Public Holiday API

Addition of a holiday column

Automated Surface Observing System (ASOS)

Addition of temperature and precipitation columns

Status of taxi stands in Daejeon

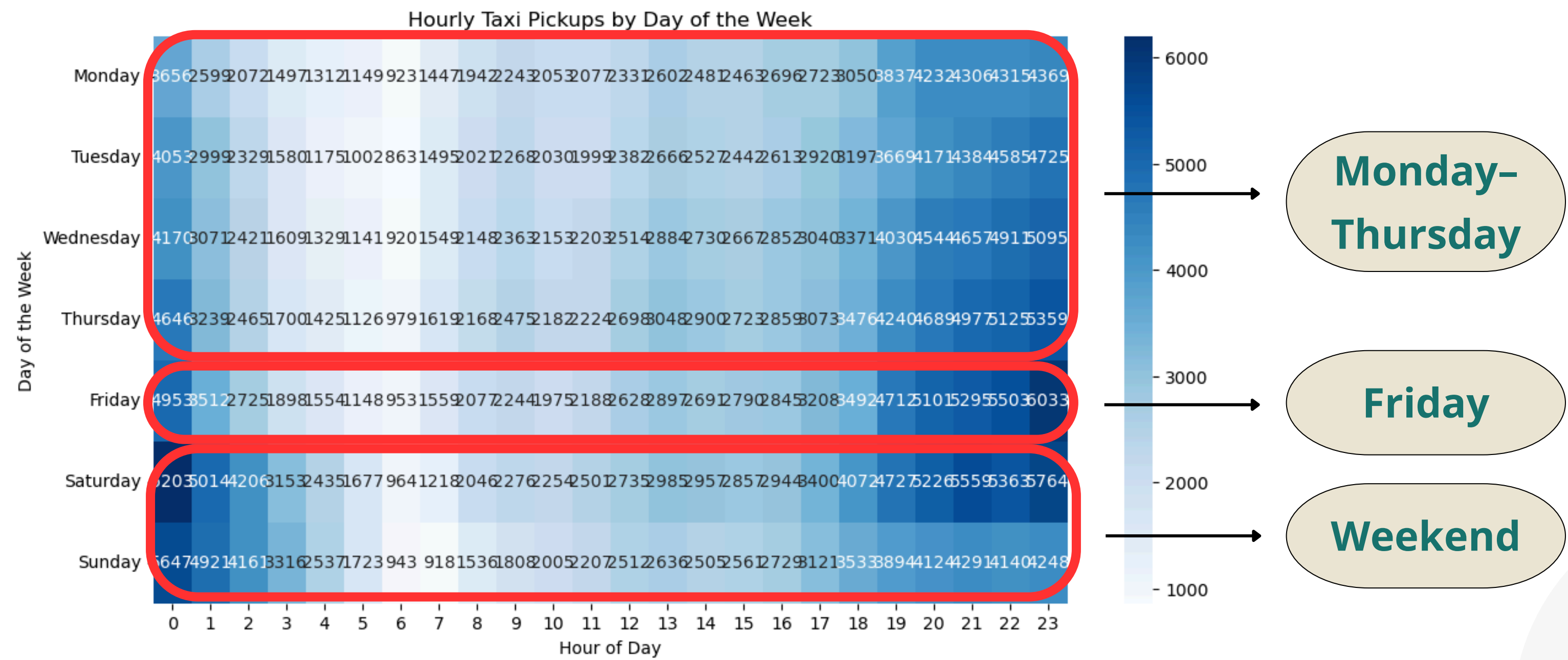
Conversion of addresses to latitude and longitude

Status of popular tourist attractions in Daejeon

Conversion of addresses to latitude and longitude

EDA – HEATMAP

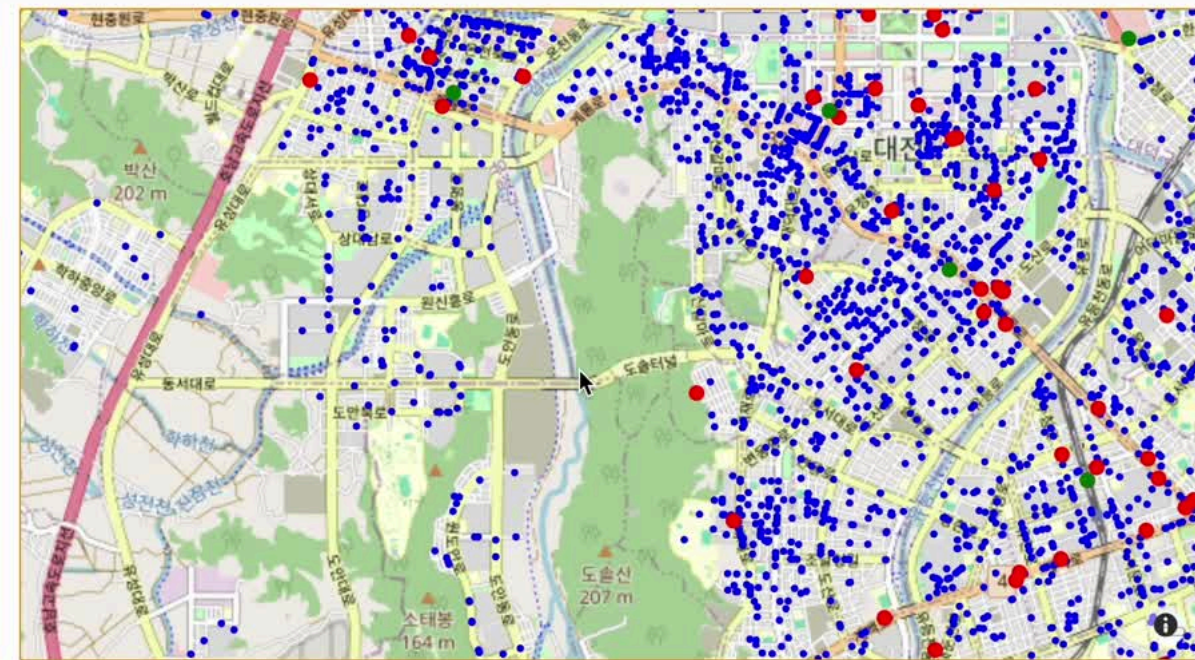
Visualization of Taxi Demand by Day of the Week and Time Period



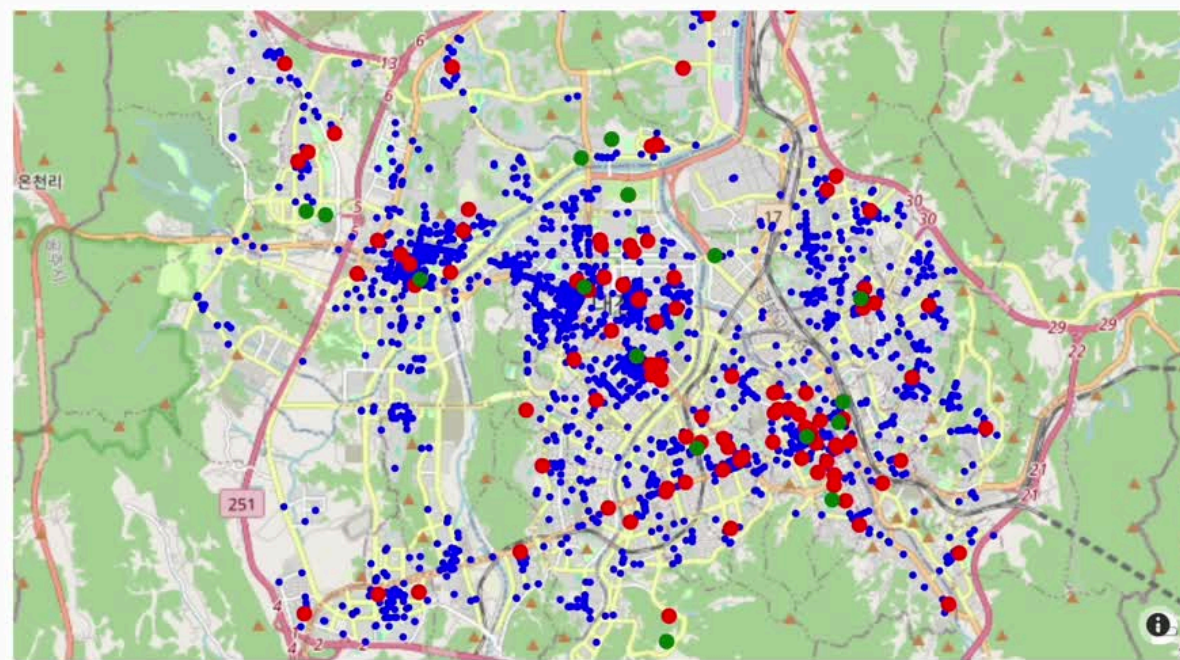
EDA – MAP VISUALIZATION

Visualization using additional data on taxi stands and popular tourist attractions

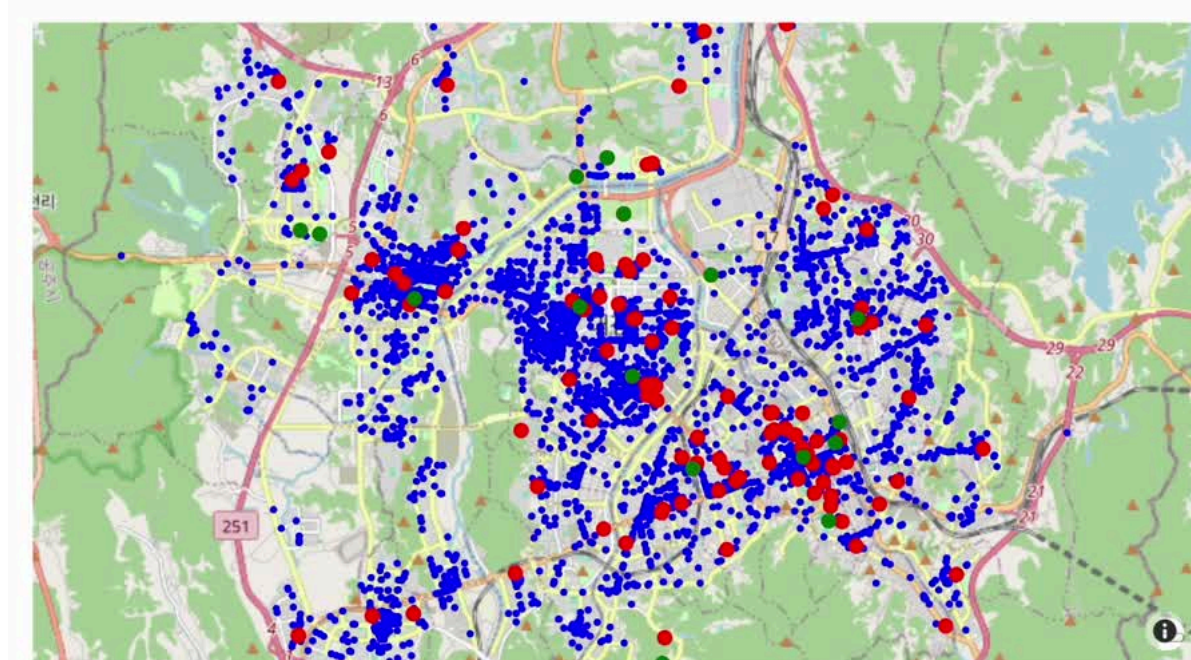
Monday–Thursday



Friday



Weekend



● Taxi Pickup
Coordinates

● Taxi Stand
Coordinates

● Popular Tourist
Attraction Coordinates

CLUSTERING

After selecting **k=3** using the elbow method and silhouette score, clustering was performed using **k-means** reflecting features such as **current location, day of the week, time, weather, and holiday status**.

Weekdays

01 Weekday chilly morning hours

02 Weekday rainy evening hours

03 Weekday holiday daytime hours

Friday

01 Friday cloudy evening hours

02 Friday chilly early morning hours

03 Friday holiday rainy noon hours

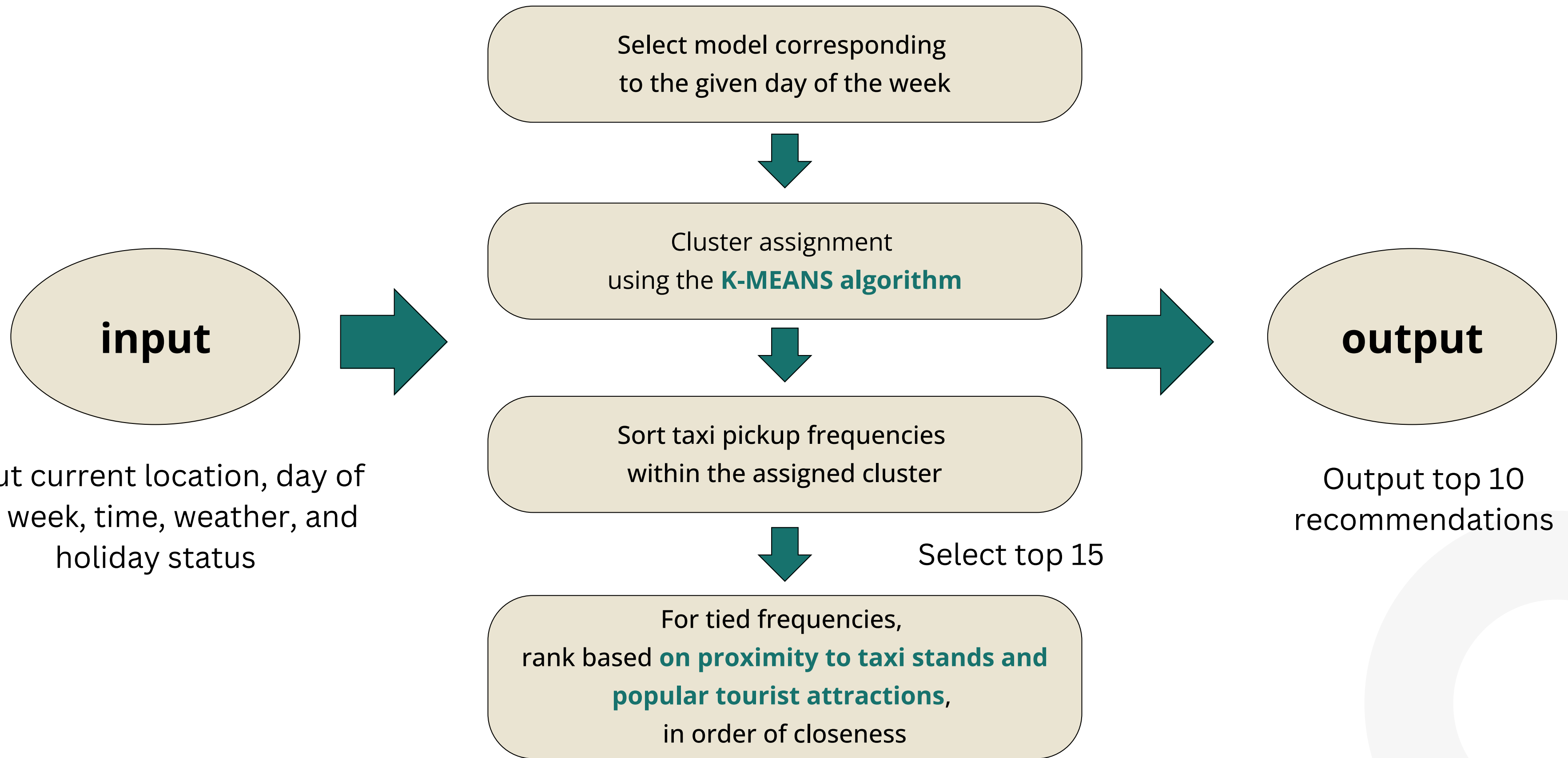
Weekend

01 Weekend chilly early morning hours

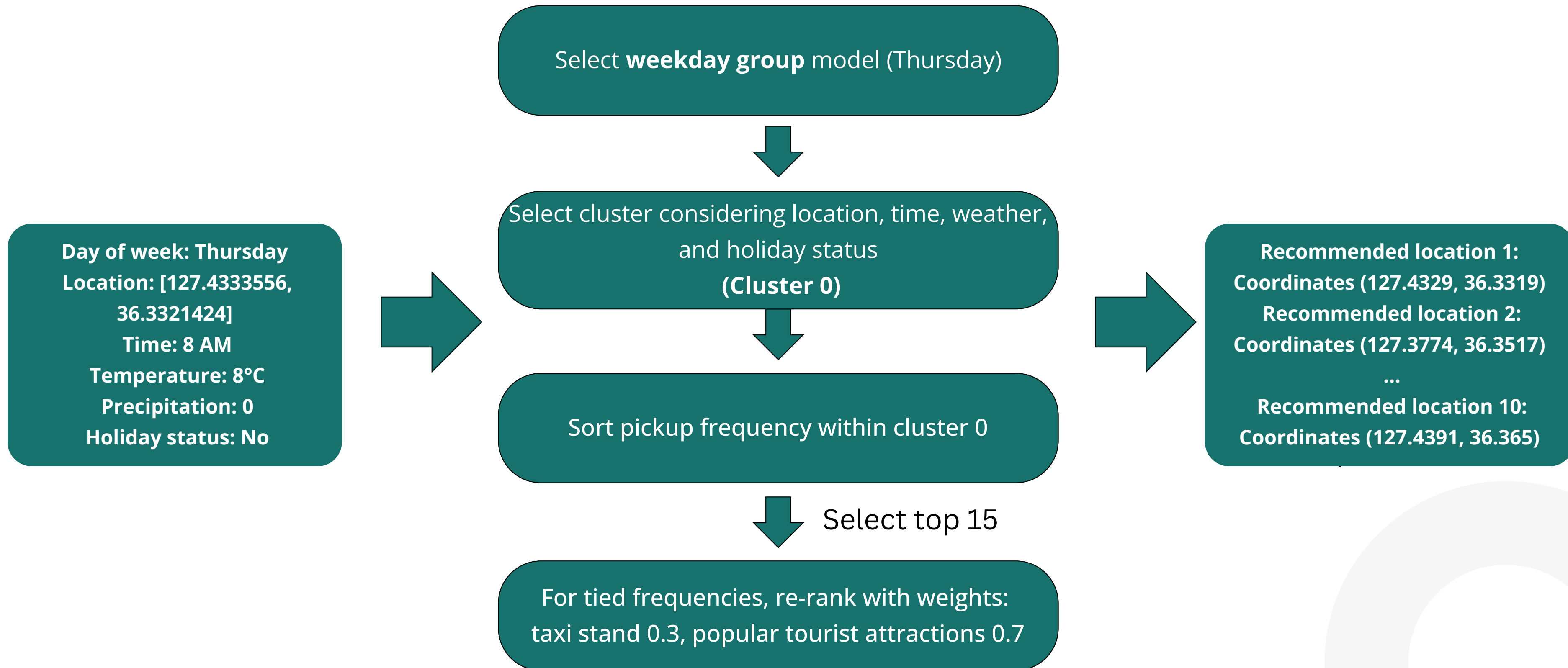
02 Weekend warm evening hours

03 Weekend holiday cold afternoon hours

RECOMMENDATION MODEL ALGORITHM



RECOMMENDATION MODEL ALGORITHM



SIMULATION

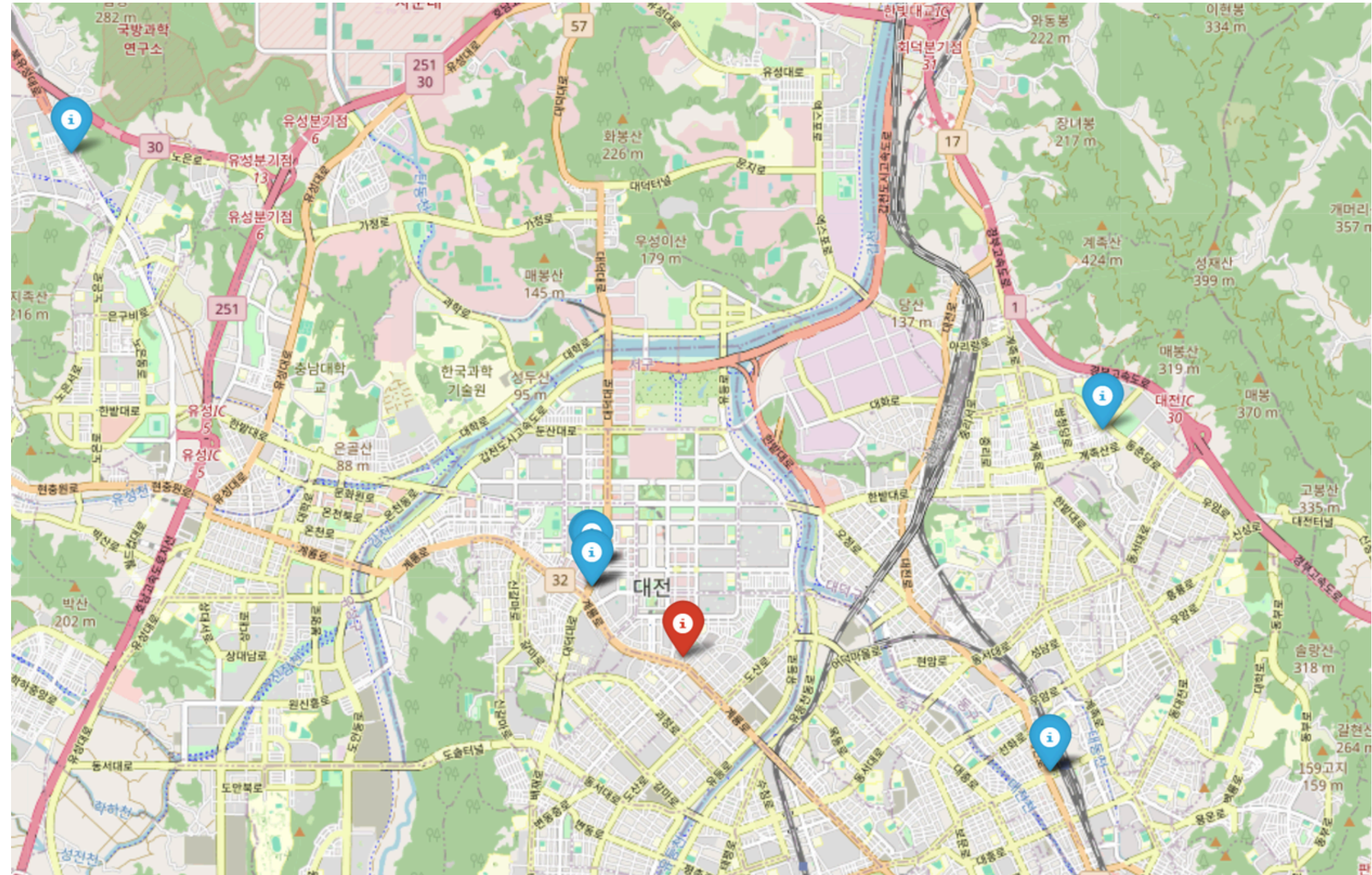
```
day = 'Thursday'
current_coors = [127.388463, 36.342952] # 예시 좌표 (경도, 위도)
current_time = 8
current_temp = 10 # 현재 기온 (예시)
current_rain = 0
is_holiday = 0

# 추천 위치 출력
top_10_locations_with_weights = recommend_locations_with_weights(
    day, current_coors, current_time, current_temp, current_rain, is_holiday,
    tourist_weight=0.8, taxi_weight=0.2
)
```

현재 상황과 유사한 승차 패턴을 추천합니다.

평일 쌀쌀한 아침 시간대 승차가 많은 곳 리스트

추천 위치 1:	좌표: (127.4329, 36.3319),	빈도: 399,	가중치 적용 점수: 0.13
추천 위치 2:	좌표: (127.3774, 36.3517),	빈도: 278,	가중치 적용 점수: 0.13
추천 위치 3:	좌표: (127.3773, 36.3517),	빈도: 276,	가중치 적용 점수: 0.13
추천 위치 4:	좌표: (127.3775, 36.3517),	빈도: 197,	가중치 적용 점수: 0.12
추천 위치 5:	좌표: (127.4328, 36.3319),	빈도: 186,	가중치 적용 점수: 0.13
추천 위치 6:	좌표: (127.3772, 36.3518),	빈도: 175,	가중치 적용 점수: 0.13
추천 위치 7:	좌표: (127.3776, 36.3499),	빈도: 139,	가중치 적용 점수: 0.28
추천 위치 8:	좌표: (127.3148, 36.3918),	빈도: 103,	가중치 적용 점수: 2.44
추천 위치 9:	좌표: (127.3775, 36.3498),	빈도: 102,	가중치 적용 점수: 0.30
추천 위치 10:	좌표: (127.4391, 36.365),	빈도: 95,	가중치 적용 점수: 1.32



● Current Location

● Recommended Location

EXPERIMENT & EVALUATION

Prepare validation dataset

24 hours

X

7 days

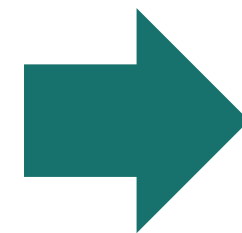
X

20 locations

= 3360 cases

For each case, generate 10 recommended locations
(33,600 rows)

Calculate **precision and recall** by considering a recommended location as correct **if there is a taxi passenger at that location during the corresponding time period of the test dataset's day group**



To check the accuracy of 10 recommended coordinates for 2 AM on Saturday, July 29, 2023:

Check how many of the 10 recommended coordinates are included among the pickup coordinates at 2 AM on Saturdays in the weekend group of the test dataset

70% accuracy: 40 cases
60% accuracy: 240 cases

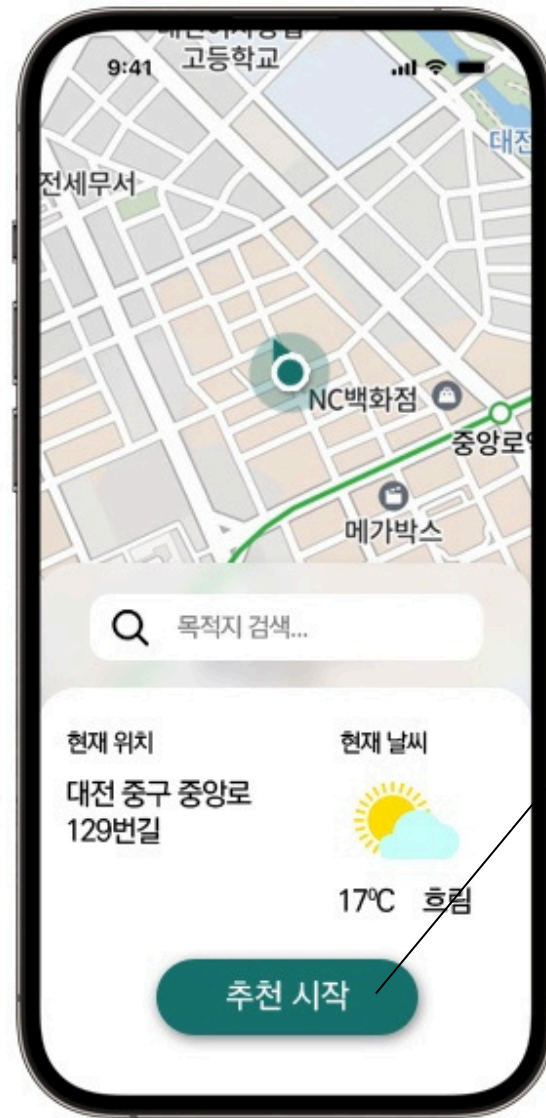
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APP INTERFACE DESIGN

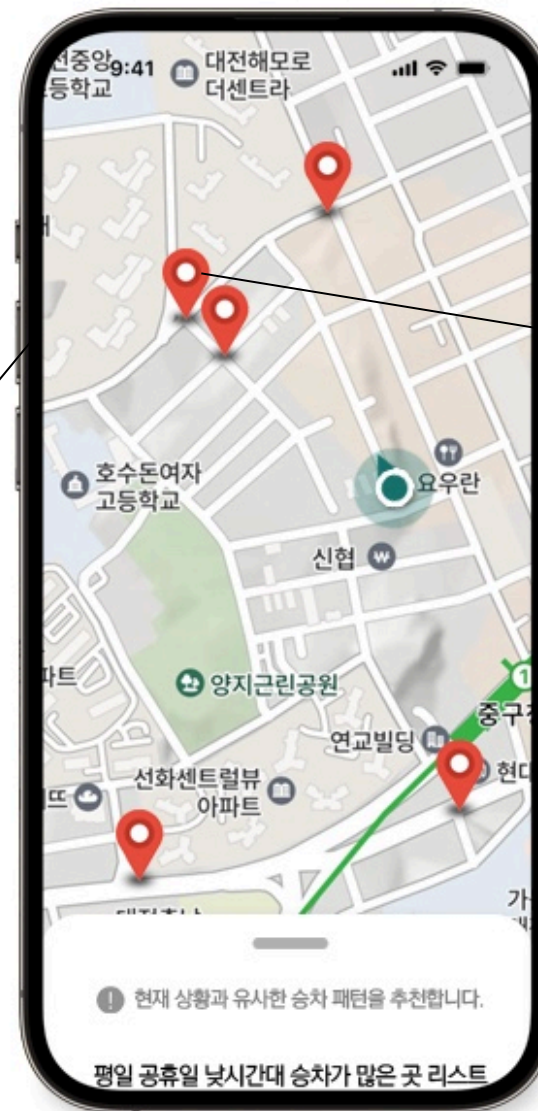


GO!CAPTAIN

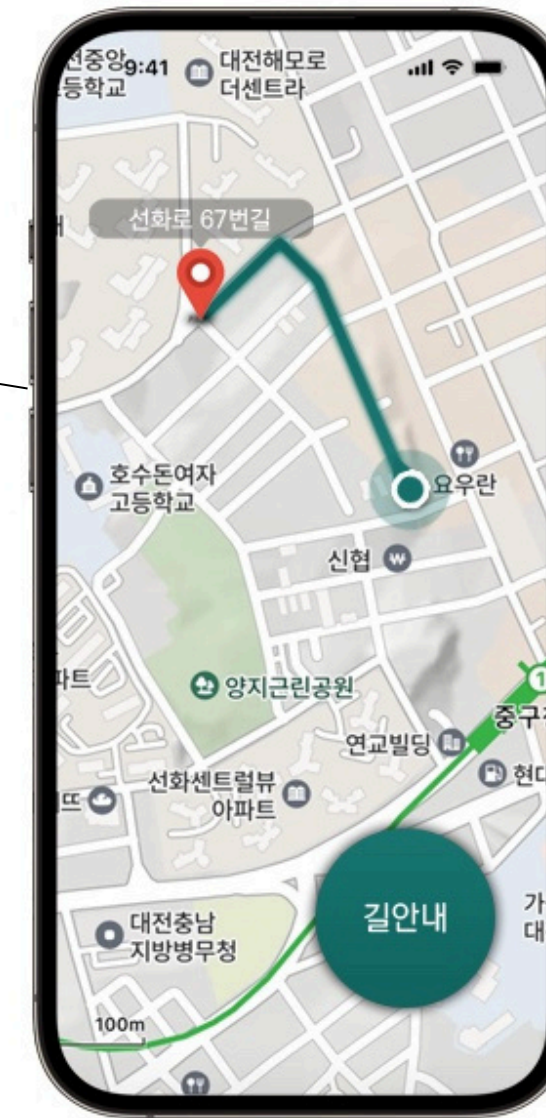
<GO! Captain Logo>



<Start Screen of the App>



<Recommended Pickup Locations Screen>



<Route Guidance Screen after Selecting a Specific Location>

IMPROVEMENT DIRECTIONS

Algorithm Structure

$$\begin{aligned} \text{weighted_score} = & (\text{min_tourist_distance} * \\ & \text{tourist_weight}) + (\text{min_taxi_distance} * \text{taxi_weight}) \\ & + (\text{new_variable_1}) * (\text{weight}) + (\text{new_variable_2}) * \\ & (\text{weight}) + \alpha \end{aligned}$$

Recommend locations by applying weights according to variable importance!

Confirmed the potential to further develop with additional variables

Driver Preference Clustering

After selecting k=3 using the elbow method and silhouette score, clustering was performed using KMeans.

→ Reflecting **pickup distance, fare, surcharge status, and driving time.**

01

Short-distance driving patterns

02

Long-distance driving and operation patterns in high-fare areas

03

Patterns of **frequent surcharge trips** with relatively **short distances**

Feature Importance:	
	Importance
할증여부	0.382649
요금	0.286235
승차거리(m)	0.183617
주행시간_초	0.147499

feature importance analysis

기대 효과

Taxi Drivers

Increased Taxi Driver Earnings

Reduced unnecessary empty driving time and generated higher income

Passengers

Improved Passenger Satisfaction

Reduced passenger waiting times through faster taxi arrivals

Reduced passenger waiting times through faster taxi arrivals

Daejeon City

Improved Traffic Flow

Enhanced taxi transportation efficiency
by providing optimized routes considering various factors



THANK YOU